



Roomies: Cohabitating Done Better

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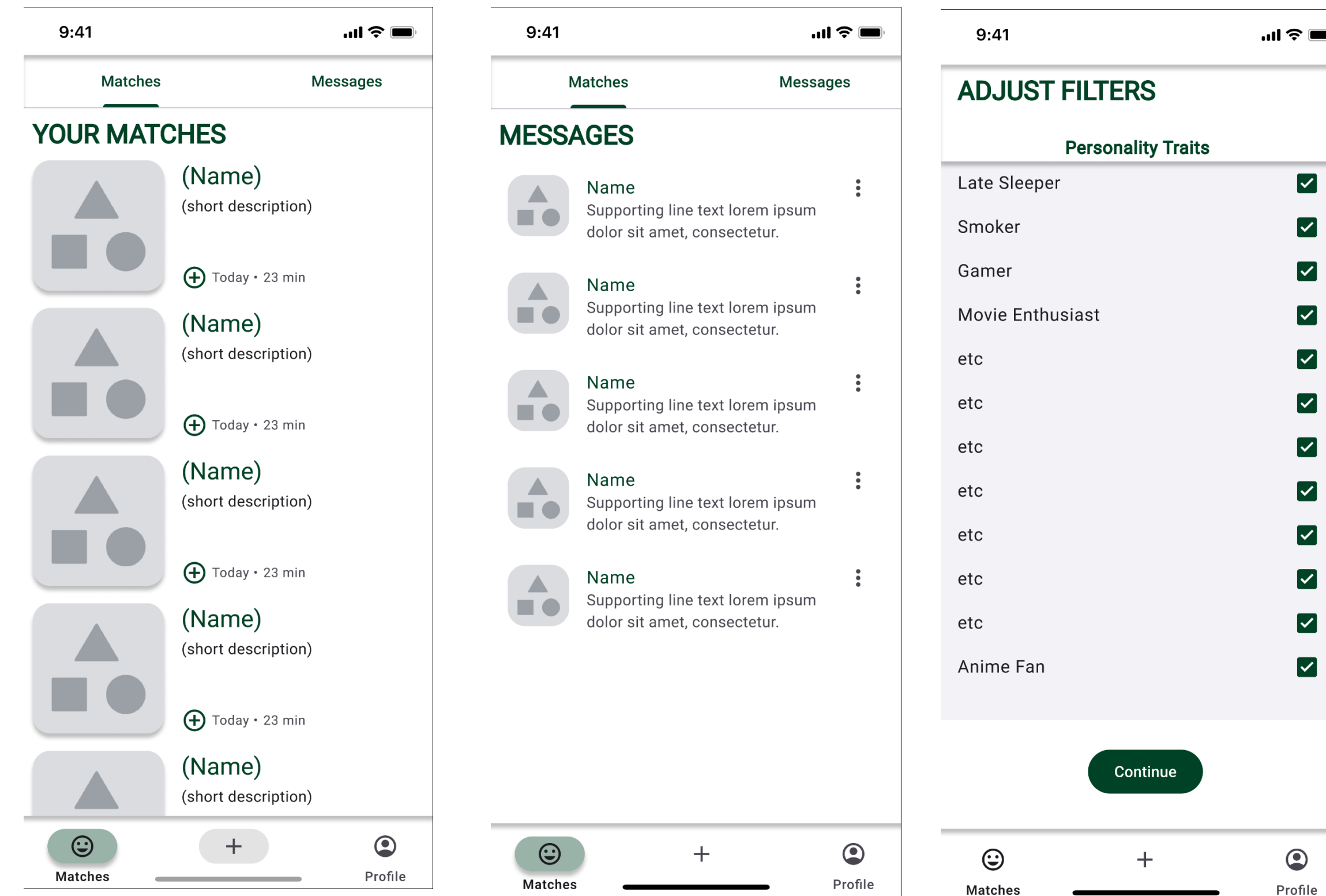
Introduction

Roomies is an app designed to make finding and living with roommates an easy and enjoyable experience. Using algorithm-based matching techniques, Roomies will pair users with the best-fitting roommates based on compatibility lifestyle habits and personality traits. Chores, groceries, and various discrepancies can all be handled via Roomie. Roomie's imbedded AI technology can even recommend the best method for handling these tasks.

Process

The creation of Roomies began with database planning, defining functional requirements, researching useful technology, and creating a budget and time scope. Following data, functionality, and research planning, the Roomies Research, Development, and Design team began to create a vision for the user interface. We followed a simple software design that is compatible with mobile devices.

Wireframes



Conclusion

Throughout this project, we learned about the detailed process of planning and designing an application. Understanding how much goes into developing software makes it very difficult to envision handling a project of this scope alone. Having a reliable and determined team is very valuable and important. Some difficulties we encountered included time scope and estimates, along with general understanding of specific requirements. It was challenging to truly grasp how much an aspect of the project required of us until we began the work. However, roomies allowed us to get more practice, and we have become better information system designers by the end of our experience.

Technology

- Not everything used is present due to space
- **Cloud Server/Hosting Services (e.g., AWS, Google Cloud):** To host backend services, store data, and manage authentication.
- **React Native or Flutter:** For building a cross-platform mobile application.
- **Figma:** For UI/UX wireframing and prototyping.
 - **Node.js / Express.js or Python (Django/Flask):** For handling server-side logic and API integration.
 - **Firebase or MongoDB:** For real-time data storage and user profile management.
- **OpenAI API:** Powers AI-based matching preferences and suggestions using a large language model (LLM).
- **Data Analytics API:** Collects and analyzes usage metrics to enhance matchmaking algorithms and platform improvements.
- **External APIs:**
- **Location Services API (e.g., Google Maps API):** Enables users to specify preferred locations and calculate distances between roommate options.